

**VIRGINIA COMMONWEALTH UNIVERSITY**

**Statistical analysis and modelling (SCMA 632)**

**A4: Multivariate Analysis and Business Analytics Applications**

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## **Introduction**

Factor Analysis (FA) is a statistical technique used to identify patterns among observed variables and reduce them to a smaller number of unobserved variables called factors.

Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms a large set of variables into a smaller set of orthogonal (uncorrelated) components.

Multidimensional Scaling (MDS) is a technique used to visualize the relative similarity or dissimilarity of individual cases in a dataset.

Conjoint Analysis is a statistical technique used to understand how consumers make trade-offs between different attributes of a product or service.

## **Objective**

The objective of Factor Analysis in the survey dataset is to uncover latent factors (like preferences or decision-making criteria) that explain the correlations among multiple observed variables related to housing preferences.

The objective of PCA on the survey dataset was to identify patterns and underlying structures within the data, potentially reducing the dimensionality for easier interpretation.

The objective of MDS on the ice cream dataset was to visualize how different ice cream brands are perceived based on multiple attributes (e.g., taste, price, availability).

In the pizza dataset, Conjoint Analysis aimed to determine the relative importance of various pizza attributes (like price, crust type, cheese amount) and their impact on consumer preferences.

# **PART 1: Principal Component Analysis & Factor Analysis**

## **INTRODUCTION**

Here, we conduct a comprehensive analysis using Factor Analysis (FA) and Principal Component Analysis (PCA) on a dataset “Survey.csv” related to housing preferences. The dataset contains various factors influencing housing decisions, such as proximity to amenities, security, pricing, and neighborhood characteristics.

## **OBJECTIVE**

### **1. Dimensionality Reduction:**

- Utilize Principal Component Analysis (PCA) to reduce the number of variables while retaining as much variance as possible.
- By identifying principal components, we aimed to simplify the dataset without losing critical information, thereby facilitating easier interpretation and analysis.

### **2. Factor Analysis:**

- Employ Factor Analysis (FA) to uncover latent variables or factors that underlie observed variables related to housing preferences.
- By examining loadings, we sought to understand which variables cluster together and contribute most significantly to each factor.

## **RESULTS & INTERPRETATIONS**

## **R**

### **1.PRINCIPAL COMPONENT ANALYSIS**

```

> names(survey_df)
[1] "City"
[3] "Age"
[5] "Monthly.Household.Income"
[7] "Planning.to.Buy.a.new.house"
[9] "Reasons.for.buying.a.house"
[11] "Number.of.rooms"
[13] "Budget"
[15] "Influence.Decision"
[17] "EMI"
[19] "X2.Proximity.to.schools"
[21] "X4..Proximity.to.work.place"
[23] "X1..Gym.Pool.Sports.facility"
[25] "X3.Power.back.up"
[27] "X5.Security"
[29] "X2..Unit.size"
[31] "X4..Layout.plan..Integrated.etc.."
[33] "X1..Price"
[35] "X3..Equated.Monthly.Instalment..EMI."
[37] "X5..Availability.of.loan"
[39] "X2..Appreciation.potential"
[41] "X4..Availability.of.domestic.help"
[43] "Size"
[45] "Maintainances"
[47] "ages"
[49] "Finished.Semi.Finished.1"

"Sex"
"Occupation"
"Income"
"Time.Frame"
"what.type.of.House"
"Size.of.House"
"Finished.Semi.Finished"
"Maintenance"
"X1.Proximity.to.city"
"X3..Proximity.to.transport"
"X5..Proximity.to.shopping"
"X2..Parking.space"
"X4.Water.supply"
"X1..Exterior.look"
"X3..Interior.design.and.branded.components"
"X5..View.from.apartment"
"X2..Booking.amount"
"X4..Maintenance.charges"
"X1..Builder.reputation"
"X3..Profile.of.neighbourhood"
"Time"
"Budgets"
"EMI.1"
"Sex"
"Influence.Decision.1"

```

```

> pca
Principal Components Analysis
Call: principal(r = sur_int, nfactors = 5, rotate = "promax", n.obs = 162)
Standardized loadings (pattern matrix) based upon correlation matrix

```

|                                            | RC1   | RC3   | RC4   | RC2   | RC5   | h2   | u2   | com |
|--------------------------------------------|-------|-------|-------|-------|-------|------|------|-----|
| X1.Proximity.to.city                       | 0.10  | 0.71  | -0.26 | 0.18  | 0.14  | 0.65 | 0.35 | 1.5 |
| X2.Proximity.to.schools                    | -0.08 | 0.57  | 0.24  | -0.11 | 0.20  | 0.51 | 0.49 | 1.8 |
| X3..Proximity.to.transport                 | -0.17 | -0.26 | 0.71  | 0.12  | 0.01  | 0.51 | 0.49 | 1.5 |
| X4..Proximity.to.work.place                | -0.38 | 0.98  | 0.03  | 0.07  | -0.10 | 0.73 | 0.27 | 1.3 |
| X5..Proximity.to.shopping                  | 0.78  | 0.12  | -0.01 | 0.27  | -0.22 | 0.69 | 0.31 | 1.5 |
| X1..Gym.Pool.Sports.facility               | 0.42  | 0.20  | 0.30  | -0.08 | -0.19 | 0.46 | 0.54 | 2.9 |
| X2..Parking.space                          | 0.53  | 0.19  | 0.04  | -0.18 | -0.06 | 0.47 | 0.53 | 1.6 |
| X3.Power.back.up                           | 0.35  | 0.53  | 0.05  | 0.09  | -0.43 | 0.59 | 0.41 | 2.8 |
| X4.Water.supply                            | 0.21  | 0.07  | 0.75  | 0.01  | 0.07  | 0.73 | 0.27 | 1.2 |
| X5.Security                                | 0.76  | -0.28 | 0.34  | -0.11 | -0.22 | 0.68 | 0.32 | 1.9 |
| X1..Exterior.look                          | 0.85  | -0.08 | -0.32 | 0.20  | 0.09  | 0.76 | 0.24 | 1.5 |
| X2..Unit.size                              | -0.06 | -0.12 | -0.04 | -0.13 | 0.74  | 0.52 | 0.48 | 1.1 |
| X3..Interior.design.and.branded.components | 0.60  | 0.22  | 0.01  | -0.07 | 0.18  | 0.66 | 0.34 | 1.5 |
| X4..Layout.plan..Integrated.etc..          | 0.34  | 0.39  | -0.05 | -0.05 | 0.32  | 0.60 | 0.40 | 3.0 |
| X5..View.from.apartment                    | 0.84  | -0.08 | 0.00  | -0.07 | 0.13  | 0.74 | 0.26 | 1.1 |
| X1..Price                                  | -0.13 | 0.16  | 0.66  | 0.13  | 0.32  | 0.53 | 0.47 | 1.8 |
| X2..Booking.amount                         | 0.22  | -0.11 | -0.08 | 0.66  | 0.04  | 0.51 | 0.49 | 1.3 |
| X3..Equated.Monthly.Instalment..EMI.       | -0.09 | -0.06 | 0.44  | 0.73  | -0.10 | 0.58 | 0.42 | 1.8 |
| X4..Maintenance.charges                    | 0.05  | -0.14 | 0.00  | 0.52  | 0.01  | 0.30 | 0.70 | 1.2 |
| X5..Availability.of.loan                   | -0.15 | 0.30  | -0.01 | 0.83  | -0.14 | 0.71 | 0.29 | 1.4 |
| X1..Builder.reputation                     | 0.08  | 0.08  | 0.34  | -0.05 | 0.68  | 0.69 | 0.31 | 1.5 |
| X2..Appreciation.potential                 | 0.23  | -0.02 | 0.13  | 0.43  | 0.41  | 0.44 | 0.56 | 2.7 |
| X3..Profile.of.neighbourhood               | 0.58  | -0.13 | 0.35  | -0.16 | 0.26  | 0.68 | 0.32 | 2.4 |
| X4..Availability.of.domestic.help          | 0.94  | -0.40 | -0.09 | -0.05 | 0.02  | 0.69 | 0.31 | 1.4 |

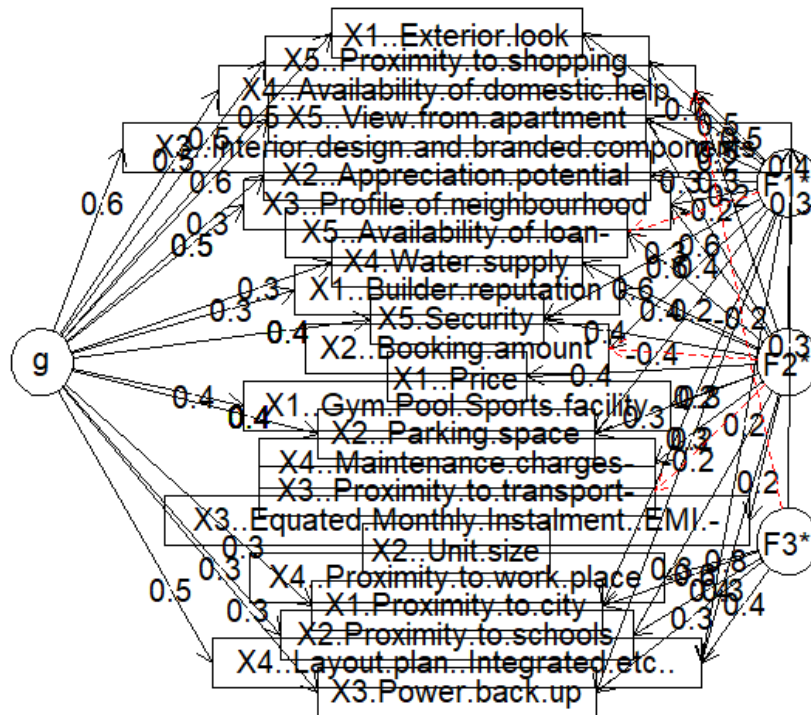
```


```

|                      | RC1  | RC3  | RC4  | RC2  | RC5  |
|----------------------|------|------|------|------|------|
| SS loadings          | 5.17 | 2.58 | 2.39 | 2.32 | 1.96 |
| Proportion Var       | 0.22 | 0.11 | 0.10 | 0.10 | 0.08 |
| Cumulative Var       | 0.22 | 0.32 | 0.42 | 0.52 | 0.60 |
| Proportion Explained | 0.36 | 0.18 | 0.17 | 0.16 | 0.14 |

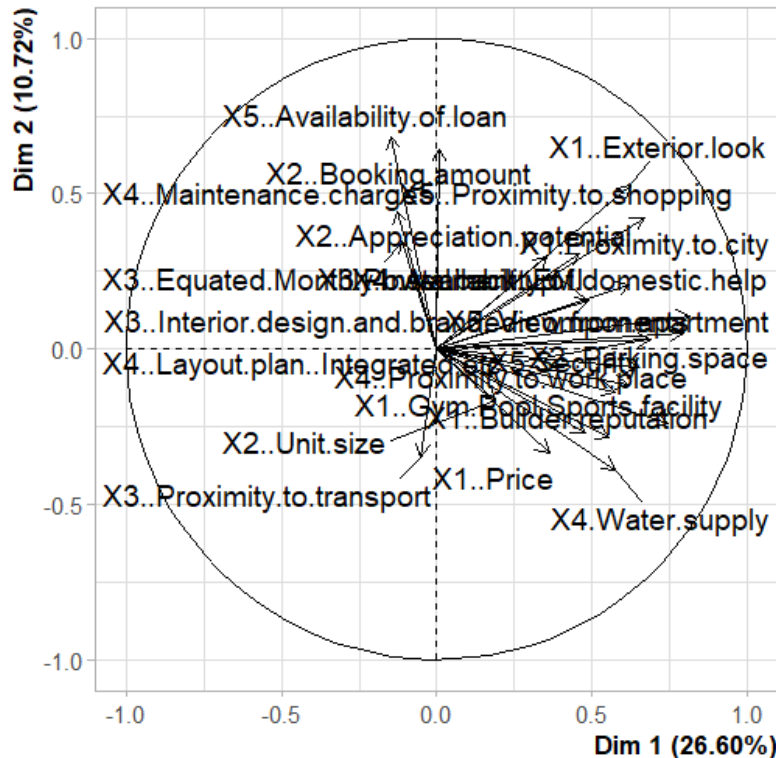


## Omega



This function(omega) is particularly useful in the context of factor analysis to understand the structure of the data in terms of general and specific factors.

## PCA graph of variables



```
> summary(pca)
```

Call:  
PCA(X = sur\_int, scale.unit = TRUE)

Eigenvalues

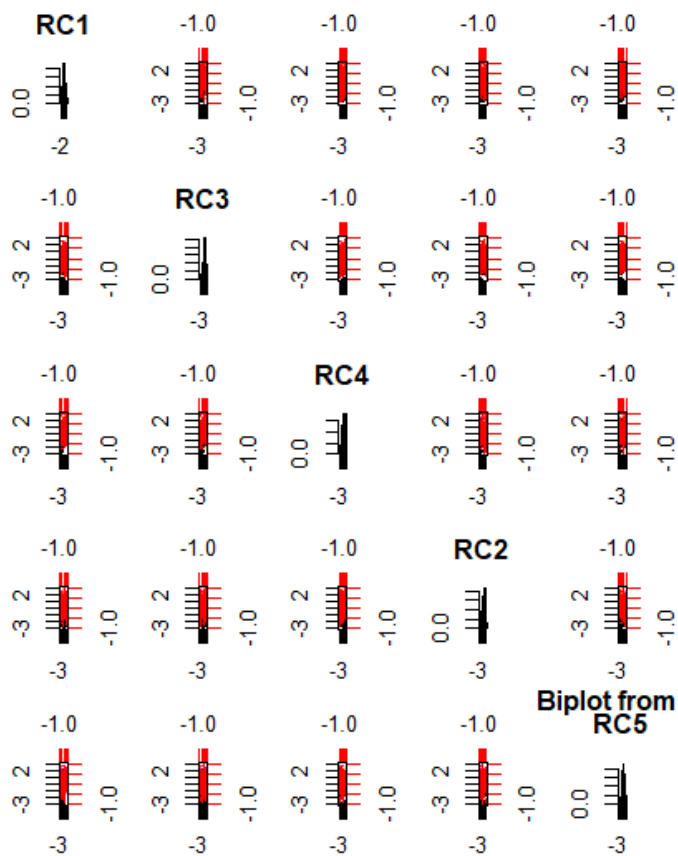
|                      | Dim.1  | Dim.2   | Dim.3  | Dim.4  | Dim.5  | Dim.6  | Dim.7  | Dim.8  | Dim.9  | Dim.10 | Dim.11 |
|----------------------|--------|---------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Variance             | 6.385  | 2.573   | 2.044  | 1.818  | 1.605  | 1.346  | 1.110  | 0.960  | 0.796  | 0.708  | 0.691  |
| % of var.            | 26.604 | 10.722  | 8.516  | 7.574  | 6.689  | 5.609  | 4.623  | 4.002  | 3.316  | 2.948  | 2.879  |
| Cumulative % of var. | 26.604 | 37.326  | 45.841 | 53.415 | 60.104 | 65.713 | 70.336 | 74.338 | 77.654 | 80.602 | 83.481 |
|                      | Dim.12 | Dim.13  | Dim.14 | Dim.15 | Dim.16 | Dim.17 | Dim.18 | Dim.19 | Dim.20 | Dim.21 | Dim.22 |
| Variance             | 0.572  | 0.510   | 0.483  | 0.444  | 0.372  | 0.312  | 0.270  | 0.243  | 0.219  | 0.206  | 0.146  |
| % of var.            | 2.383  | 2.126   | 2.013  | 1.849  | 1.552  | 1.301  | 1.126  | 1.013  | 0.913  | 0.859  | 0.609  |
| Cumulative % of var. | 85.864 | 87.990  | 90.003 | 91.852 | 93.404 | 94.705 | 95.831 | 96.844 | 97.757 | 98.616 | 99.226 |
|                      | Dim.23 | Dim.24  |        |        |        |        |        |        |        |        |        |
| Variance             | 0.107  | 0.079   |        |        |        |        |        |        |        |        |        |
| % of var.            | 0.445  | 0.329   |        |        |        |        |        |        |        |        |        |
| Cumulative % of var. | 99.671 | 100.000 |        |        |        |        |        |        |        |        |        |

Individuals (the 10 first)

|    | Dist  | Dim.1  | ctr   | cos2  | Dim.2  | ctr   | cos2  | Dim.3  |
|----|-------|--------|-------|-------|--------|-------|-------|--------|
| 1  | 6.300 | -0.252 | 0.014 | 0.002 | -3.818 | 8.092 | 0.367 | 0.313  |
| 2  | 6.395 | -3.145 | 2.213 | 0.242 | -3.264 | 5.914 | 0.260 | 0.136  |
| 3  | 7.295 | -3.349 | 2.509 | 0.211 | -3.248 | 5.855 | 0.198 | 4.694  |
| 4  | 7.180 | 5.892  | 7.768 | 0.673 | -2.340 | 3.040 | 0.106 | -0.752 |
| 5  | 4.219 | -0.112 | 0.003 | 0.001 | 0.173  | 0.017 | 0.002 | -0.419 |
| 6  | 4.059 | -0.563 | 0.071 | 0.019 | -1.617 | 1.452 | 0.159 | -0.048 |
| 7  | 5.179 | 3.235  | 2.342 | 0.390 | -0.094 | 0.005 | 0.000 | -0.176 |
| 8  | 6.996 | -5.449 | 6.643 | 0.607 | -1.946 | 2.102 | 0.077 | -0.121 |
| 9  | 4.453 | -2.765 | 1.711 | 0.386 | -1.244 | 0.859 | 0.078 | -1.072 |
| 10 | 4.841 | -0.266 | 0.016 | 0.003 | -0.129 | 0.009 | 0.001 | 0.232  |

```
biplot(pca, scale = 0)
```





|                              | ctr    | cos2  |       |  |
|------------------------------|--------|-------|-------|--|
| 1                            | 0.068  | 0.002 |       |  |
| 2                            | 0.013  | 0.000 |       |  |
| 3                            | 15.401 | 0.414 |       |  |
| 4                            | 0.396  | 0.011 |       |  |
| 5                            | 0.123  | 0.010 |       |  |
| 6                            | 0.002  | 0.000 |       |  |
| 7                            | 0.022  | 0.001 |       |  |
| 8                            | 0.010  | 0.000 |       |  |
| 9                            | 0.803  | 0.058 |       |  |
| 10                           | 0.038  | 0.002 |       |  |
| Variables (the 10 first)     |        |       |       |  |
|                              | Dim.1  | ctr   | cos2  |  |
| X1.Proximity.to.city         | 0.460  | 3.316 | 0.212 |  |
| X2.Proximity.to.schools      | 0.478  | 3.585 | 0.229 |  |
| X3..Proximity.to.transport   | -0.050 | 0.040 | 0.003 |  |
| X4..Proximity.to.work.place  | 0.208  | 0.679 | 0.043 |  |
| X5..Proximity.to.shopping    | 0.667  | 6.975 | 0.445 |  |
| X1..Gym.Pool.Sports.facility | 0.576  | 5.187 | 0.331 |  |
| X2..Parking.space            | 0.647  | 6.554 | 0.418 |  |
| X3.Power.back.up             | 0.489  | 3.739 | 0.239 |  |
| X4.Water.supply              | 0.576  | 5.194 | 0.332 |  |
| X5.Security                  | 0.617  | 5.959 | 0.380 |  |
| X1.Proximity.to.city         |        |       |       |  |
| X2.Proximity.to.schools      |        |       |       |  |
| X3..Proximity.to.transport   |        |       |       |  |
| X4..Proximity.to.work.place  |        |       |       |  |
| X5..Proximity.to.shopping    |        |       |       |  |
| X1..Gym.Pool.Sports.facility |        |       |       |  |
| X2..Parking.space            |        |       |       |  |
| X3.Power.back.up             |        |       |       |  |
| X4.Water.supply              |        |       |       |  |
| X5.Security                  |        |       |       |  |

Interpretation:

- **First Component (RC1):**
  - High loadings on variables related to proximity to shopping, security, exterior look, view from the apartment, and profile of the neighborhood. This suggests that RC1 captures aspects related to location convenience and aesthetic appeal.
- **Second Component (RC2):**
  - High loadings on variables related to booking amount, EMI, availability of loans, and maintenance charges. This component appears to capture financial aspects and affordability.
- **Third Component (RC3):**
  - High loadings on variables related to proximity to the city and schools. This component captures aspects related to accessibility and proximity to essential services.
- **Fourth Component (RC4):**
  - High loadings on proximity to transport, water supply, and price. This component captures aspects related to basic amenities and cost.
- **Fifth Component (RC5):**
  - High loadings on unit size and builder reputation. This component captures aspects related to the size and reputation of the property.

## 2.FACTOR ANALYSIS

```
> names(factor_analysis)
[1] "residual"      "dof"           "chi"           "nh"            "rms"           "EPVAL"
[7] "crms"          "EBIC"          "ESABIC"        "fit"           "fit.off"       "sd"
[13] "factors"       "complexity"    "n.obs"         "objective"     "criteria"      "STATISTIC"
[19] "PVAL"         "Call"         "null.model"    "null.dof"      "null.chisq"    "TLI"
[25] "CFI"          "RMSEA"        "BIC"           "SABIC"         "r.scores"      "R2"
[31] "valid"        "score.cor"     "weights"       "rotation"      "hyperplane"    "communality"
[37] "communalities" "uniquenesses" "values"        "e.values"      "loadings"      "model"
[43] "fm"           "rot.mat"      "Structure"     "method"        "scores"        "R2.scores"
[49] "r"            "np.obs"       "fn"            "Vaccounted"    "ECV"
```

```
> print(factor_analysis$loadings,reorder=TRUE)

Loadings:

```

|                                            | MR1    | MR3    | MR4    | MR2    |
|--------------------------------------------|--------|--------|--------|--------|
| X1.Proximity.to.city                       | 0.338  | 0.719  | -0.170 |        |
| X2.Proximity.to.schools                    | 0.222  | 0.406  | 0.258  | -0.186 |
| X3..Proximity.to.transport                 | -0.121 | -0.225 | 0.501  |        |
| X4..Proximity.to.work.place                |        | 0.615  |        |        |
| X5..Proximity.to.shopping                  | 0.652  | 0.113  |        | 0.251  |
| X1..Gym.Pool.Sports.facility               | 0.426  | 0.149  | 0.223  | -0.139 |
| X2..Parking.space                          | 0.529  | 0.184  |        | -0.161 |
| X3.Power.back.up                           | 0.356  | 0.303  |        |        |
| X4.Water.supply                            | 0.382  |        | 0.752  |        |
| X5.Security                                | 0.594  | -0.195 | 0.275  |        |
| X1..Exterior.look                          | 0.783  |        | -0.268 | 0.285  |
| X2..Unit.size                              | 0.129  |        |        | -0.132 |
| X3..Interior.design.and.branded.components | 0.691  | 0.282  |        |        |
| X4..Layout.plan..Integrated.etc..          | -0.539 | 0.429  |        | -0.125 |
| X5..View.from.apartment                    | 0.828  |        |        |        |
| X1..Price                                  | 0.154  | 0.201  | 0.532  |        |
| X2..Booking.amount                         | 0.108  |        | -0.140 | 0.519  |
| X3..Equated.Monthly.Instalment..EMI.       | -0.101 |        | 0.218  | 0.520  |
| X4..Maintenance.charges                    |        |        | -0.108 | 0.318  |
| X5..Availability.of.loan                   | -0.150 | 0.239  |        | 0.878  |
| X1..Builder.reputation                     | 0.388  | 0.254  | 0.386  | -0.180 |
| X2..Appreciation.potential                 | 0.309  | 0.118  |        | 0.234  |
| X3..Profile.of.neighbourhood               | 0.658  |        | 0.380  | -0.212 |
| X4..Availability.of.domestic.help          | 0.768  | -0.213 |        |        |
| Time                                       | 0.118  |        |        | 0.370  |
| Size                                       | 0.713  | 0.383  | 0.257  |        |
| Budgets                                    | 0.709  | 0.400  | 0.299  |        |
| Maintainances                              | 0.707  | 0.405  | 0.322  |        |
| EMI.1                                      | 0.700  | 0.508  | 0.214  | -0.113 |

|                | MR1   | MR3   | MR4   | MR2   |
|----------------|-------|-------|-------|-------|
| SS loadings    | 7.135 | 2.559 | 2.113 | 1.994 |
| Proportion Var | 0.246 | 0.088 | 0.073 | 0.069 |
| Cumulative Var | 0.246 | 0.334 | 0.407 | 0.476 |

The loadings indicate how much each variable contributes to the identified factors (MR1, MR2, MR3, MR4). The factors can be interpreted based on the variables that load highly on them.

- **MR1 (Factor 1):**

This factor seems to capture aspects related to the aesthetic appeal and comprehensive features of the property, including proximity to shopping, design, neighborhood profile, and overall size and budget considerations.

- **MR2 (Factor 2):**

This factor captures financial and logistical aspects related to acquiring the property, such as loan availability, booking amount, and EMI.

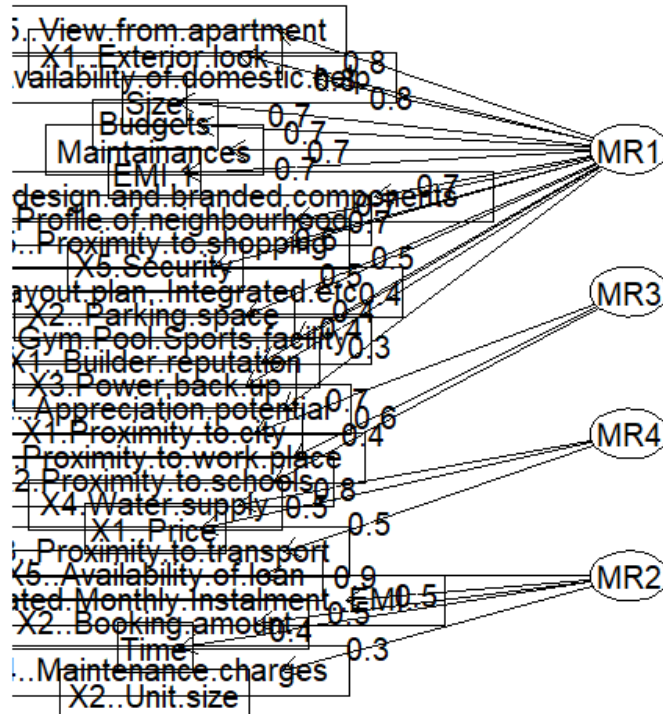
- **MR3 (Factor 3):**

This factor seems to capture accessibility aspects such as proximity to city, schools, shopping, and availability of amenities like parking and power backup.

- **MR4 (Factor 4):**

This factor appears to capture essential services and affordability aspects, such as proximity to transport, water supply, and price considerations.

### Factor Analysis



```
print(factor_analysis$communality)
X1..Proximity.to.city          0.66641636
X2..Proximity.to.schools      0.31505092
X3..Proximity.to.transport    0.32255808
X4..Proximity.to.work.place   0.38704235
X5..Proximity.to.shopping     0.50168188
X1..Gym.Pool.Sports.facility  0.27313215
X2..Parking.space            0.34504278
X3..Power.back.up            0.21911638
X4..Water.supply              0.72104641
X5.Security                   0.47189943
X1..Exterior.look            0.77466504
X2..Unit.size                 0.04040037
X3..Interior.design.and.branded.components 0.57000009
X4..Layout.plan..Integrated.etc.. 0.49408411
X5..View.from.apartment      0.69182220
X1..Price                     0.35455614
X2..Booking.amount           0.30237766
X3..Equated.Monthly.Instalment..EMI. 0.33109089
X4..Maintenance.charges      0.12587052
X5..Availability.of.loan      0.85702785
X1..Builder.reputation        0.39708855
X2..Appreciation.potential    0.17201887
X3..Profile.of.neighbourhood  0.62294275
X4..Availability.of.domestic.help 0.63983787
Time                           0.15522805
Size                           0.72147280
Budgets                       0.75210691
Maintainances                  0.76752409
```

EMI.1  
0.80701346

```
> print(factor_analysis$scores)
      MR1      MR3      MR4      MR2
[1,] -0.5876607 -0.11251823 2.050119670 -0.80909066
[2,] -1.4792705 -0.36503965 0.775643390 -1.16305313
[3,] -0.5478608 -3.35305405 0.945557247 -1.32923887
[4,] 1.8807274 0.13402370 0.266228876 -2.06159338
[5,] 0.3295689 -0.18023376 -1.170876218 -0.36010367
[6,] -0.2866020 -0.47379445 0.367382156 -0.21010883
[7,] 0.5003641 0.65274995 1.058454876 0.43822528
[8,] -1.9088343 -0.71638354 0.125781819 -1.29105206
[9,] -0.7069313 0.10856075 -0.040407655 0.37696557
[10,] 0.1573651 0.97898633 -0.458207894 0.32151704
[11,] -0.3560346 -0.87215158 -0.548893781 0.20034247
[12,] -1.0750688 1.06862183 -1.324450733 -0.87149108
[13,] -1.2632780 0.08817647 0.497275087 0.87321384
[14,] -0.1478539 0.79809053 0.072043274 0.84550179
[15,] -0.8943020 -0.57613016 0.146791486 -0.06867479
[16,] -2.3861008 -0.26161531 -0.224930358 0.28216534
[17,] -1.6756359 0.98678140 -0.553005149 -0.12911862
[18,] -1.3558992 1.21940046 0.959335531 -0.38269031
[19,] 0.3070668 1.15207947 0.709196598 -1.63990272
[20,] 0.6265597 -0.78100996 0.357780784 -2.31140305
[21,] -1.3558992 1.21940046 0.959335531 -0.38269031
[22,] -0.4194756 1.30069137 0.503442662 0.91390513
[23,] -1.1801257 1.51209556 1.086366589 -0.54903222
[24,] 0.5281885 -0.61791805 -0.458074607 0.95748245
[25,] 1.5544448 0.47387769 0.923143557 0.02046889
[26,] -0.6560738 1.28545333 -0.217972896 0.05690345
[27,] -0.5592703 0.09090353 -0.895706580 1.14485399
[28,] -0.2026950 0.43630856 0.148534968 0.28641784
[29,] -1.3721154 0.67538936 0.895056459 -0.08885140
[30,] 1.0626369 -0.05059363 -0.642613792 -0.85437235
[31,] 0.8055042 0.64353368 0.081147945 -0.65568444
[32,] -0.6404405 -0.24517693 -1.178682910 -0.09102854
[33,] -0.5394157 -0.09824719 0.202530097 1.22865673
[34,] 0.8810889 -0.47378272 0.659986298 0.70591671
[35,] -1.4194026 -0.95544542 0.515394129 0.77098383
[36,] 0.4944889 -0.80617264 0.926183237 -0.84615059
[37,] -0.3920544 0.11286019 -0.185478395 1.25609547
[38,] 0.9187362 0.37784351 -2.021924632 0.03073602
[39,] -0.2688068 0.43165531 0.135943047 0.71033819
[40,] 0.3238506 -1.31842216 -0.956078235 -0.04785917
[41,] -0.6013308 0.39092682 -1.170649492 -0.39382993
[42,] 0.7672588 -0.97998184 1.065783179 1.38843255
[43,] 1.2268022 0.92424961 -0.506390529 -1.74468234
[44,] 0.1531872 -0.77891730 0.598245535 1.48113689
[45,] 0.0310595 -1.91707747 -0.829528716 1.67981703
[46,] 1.5259121 0.37706979 -0.106363781 0.67027212
[47,] 0.1309902 -0.37026373 -2.097303080 -0.58011507
[48,] 0.6094975 -0.49059484 -1.347669950 0.20423233
[49,] 1.7828850 1.41737185 0.926221436 1.50375524
[50,] -1.3382877 -0.75161725 -2.343753911 -1.08189200
[51,] -0.4664299 -2.63297854 0.768856976 -0.96279188
[52,] 1.6302221 0.38795119 0.696176766 -0.46016474
```

```
[53,] 0.3446095 -0.26406859 0.785862102 1.91000860
[54,] 1.3756285 0.60316568 0.936286214 -0.57622139
[55,] 0.7398498 0.67345244 0.169092222 0.57967423
[56,] 0.3269372 0.06586277 -1.542789689 -0.16210359
[57,] -0.1855857 0.17245157 0.001009407 1.08219628
[58,] 1.8807274 0.13402370 0.266228876 -2.06159338
[59,] 0.9987618 1.15565927 0.692293241 1.45269523
[60,] 0.2569674 -0.23430610 0.186779172 1.47210427
[61,] 0.5694039 -0.35107186 1.166005563 -0.37318649
[62,] 0.8810067 -0.59364313 -0.707466321 -0.20736685
[63,] 0.9555754 0.24108053 -1.713145455 0.32223769
[64,] 0.2601443 -1.07529756 0.502800902 -1.87754200
[65,] -0.8113446 0.98765809 0.277402326 -0.20207944
[66,] -0.1096387 -1.38076313 0.216101775 1.88705736
[67,] 0.8768180 -0.28471376 -1.360191871 0.28202914
[68,] -0.6013308 0.39092682 -1.170649492 -0.39382993
[69,] -0.4169898 0.74655193 -0.353765544 -0.60564613
[70,] 0.5132101 -0.05290097 1.503170655 0.48989636
```

## Interpretation of Proportions:

- **MR1** explains the largest portion of the variance (24.6%), indicating that factors related to aesthetic appeal and comprehensive property features are the most significant in this survey.
- **MR2, MR3, and MR4** explain smaller proportions of the variance but still contribute important dimensions related to financial/logistical aspects, accessibility, and essential services/affordability, respectively.

## PYTHON

### 1.PCA

```
Explained variance ratio: [0.31822766 0.09526344 0.06889323 0.06341429 0.06111318]
Cumulative explained variance: [0.31822766 0.4134911 0.48238432 0.54579862 0.6069118 ]
```

## Explained Variance Ratio:

- This is a measure of how much variance each principal component explains from the total variance in the dataset.

For example,

- PC1 explains approximately 31.82% of the total variance in the data.
- PC2 explains an additional 9.53%.

Together, these ratios provide insight into how much information each principal component carries. A higher ratio indicates that the principal component explains a larger portion of the variance in the original data.

## Cumulative Explained Variance:

- This shows the cumulative proportion of variance explained by the principal components.
- For example:
  - The first two principal components together explain approximately 41.35% (31.82% + 9.53%) of the total variance.
  - The first three explain 48.24%, and so forth.

This cumulative measure helps determine how many principal components are needed to explain a significant portion of the data's variance. It's useful for deciding how many principal components to retain in the analysis.

#### PCA Components:

|   | 3. Proximity to transport | 4. Proximity to work place \ |
|---|---------------------------|------------------------------|
| 0 | 0.014928                  | -0.052653                    |
| 1 | 0.126411                  | 0.060153                     |
| 2 | -0.465915                 | 0.052699                     |
| 3 | 0.069723                  | -0.592034                    |
| 4 | 0.271472                  | -0.097866                    |

|   | 5. Proximity to shopping | 1. Gym/Pool/Sports facility | 2. Parking space \ |
|---|--------------------------|-----------------------------|--------------------|
| 0 | -0.210212                | -0.180419                   | -0.203748          |
| 1 | -0.270909                | 0.089917                    | 0.037004           |
| 2 | 0.129612                 | 0.015402                    | 0.146867           |
| 3 | -0.053158                | -0.081124                   | -0.056974          |
| 4 | 0.177992                 | 0.292307                    | 0.182483           |

|   | 3. Power back-up | 4. Water supply | 5. Security | 1. Exterior look \ |
|---|------------------|-----------------|-------------|--------------------|
| 0 | -0.155294        | -0.203859       | -0.196652   | -0.219630          |
| 1 | -0.076559        | 0.163703        | 0.022475    | -0.316341          |
| 2 | 0.156561         | -0.339980       | -0.003218   | 0.177654           |
| 3 | -0.426886        | -0.075773       | 0.191186    | 0.138517           |
| 4 | 0.189252         | 0.209854        | 0.454456    | -0.093192          |

|   | 2. Unit size ... | 5. Availability of loan | 1. Builder reputation \ |
|---|------------------|-------------------------|-------------------------|
| 0 | -0.054351 ...    | 0.050746                | -0.198477               |
| 1 | 0.088531 ...     | -0.465934               | 0.182091                |
| 2 | -0.023657 ...    | -0.216143               | -0.217037               |
| 3 | 0.330608 ...     | -0.221261               | 0.077041                |
| 4 | -0.365054 ...    | -0.073568               | -0.298374               |

|   | 2. Appreciation potential | 3. Profile of neighbourhood \ |
|---|---------------------------|-------------------------------|
| 0 | -0.118254                 | -0.253555                     |
| 1 | -0.190878                 | 0.132598                      |
| 2 | -0.199931                 | -0.080191                     |
| 3 | 0.123135                  | 0.155028                      |
| 4 | -0.148766                 | 0.093081                      |

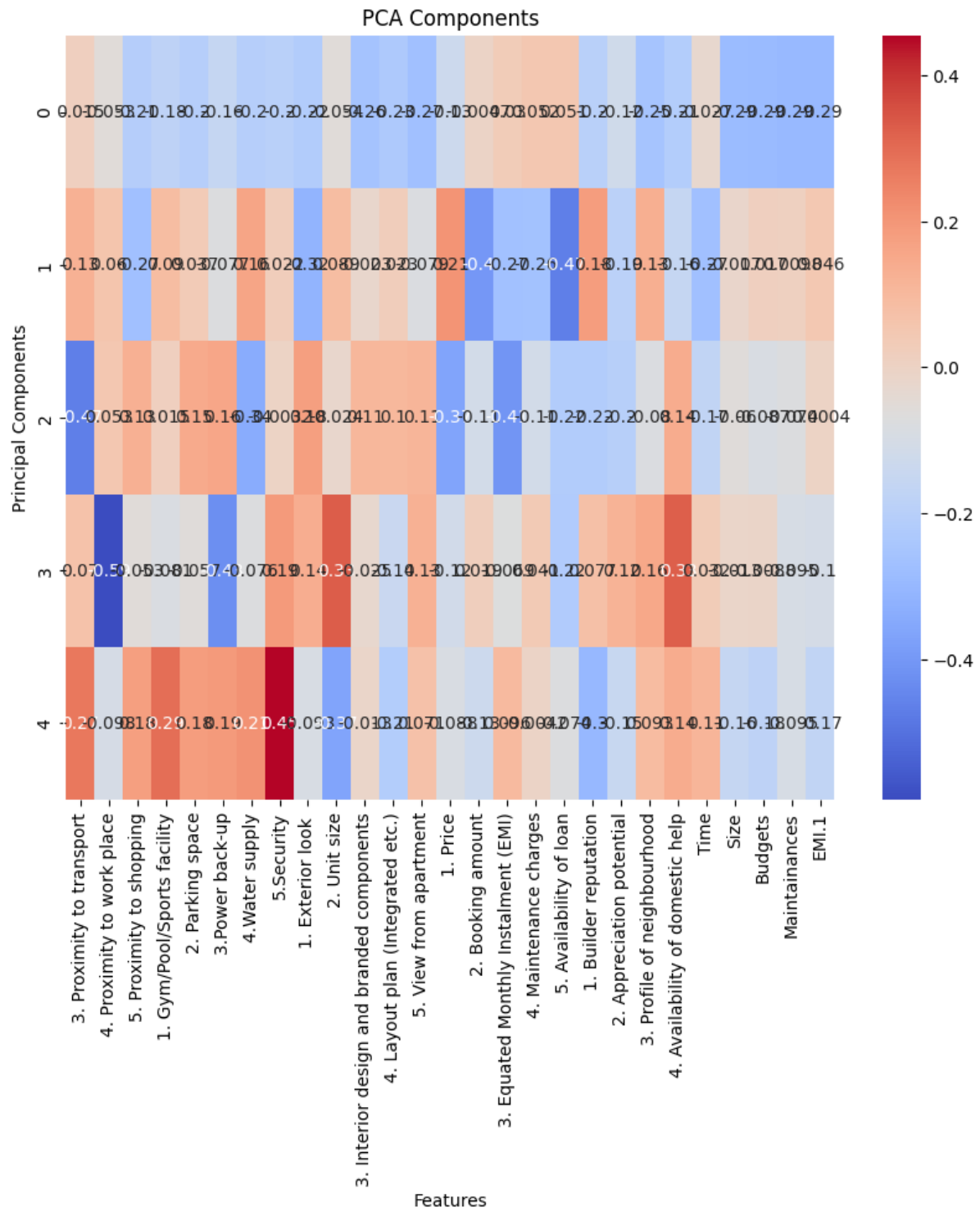
|   | 4. Availability of domestic help | Time      | Size      | Budgets \ |
|---|----------------------------------|-----------|-----------|-----------|
| 0 | -0.213143                        | -0.026919 | -0.288181 | -0.292770 |
| 1 | -0.156897                        | -0.270334 | -0.016512 | 0.016734  |
| 2 | 0.143303                         | -0.170545 | -0.060315 | -0.086750 |
| 3 | 0.325950                         | 0.032492  | -0.013361 | -0.008785 |
| 4 | 0.142015                         | 0.114240  | -0.155017 | -0.181846 |

|   | Maintainances | EMI.1     |
|---|---------------|-----------|
| 0 | -0.294955     | -0.294392 |
| 1 | 0.009794      | 0.046093  |
| 2 | -0.074044     | 0.000399  |
| 3 | -0.095202     | -0.101849 |
| 4 | -0.095451     | -0.167535 |

[5 rows x 27 columns]

The PCA components matrix represents the contribution of each original variable to the principal components. Each row corresponds to a principal component, and each column represents a feature from the dataset. The values in the matrix indicate the weights (or loadings) of each feature in the corresponding principal component.





## 2.FA

Factor Variance:

```
(array([4.97928835, 3.6638373, 1.99739684, 1.46935161, 1.31270193]), array([0.18441809, 0.13569768, 0.07397766, 0.05442043, 0.04861859]), array([0.18441809, 0.32011576, 0.39409343, 0.44851386, 0.49713245]))
```

### Eigenvalues (Array 1):

- These are the eigenvalues associated with each factor extracted from the factor analysis.
- Eigenvalues represent the amount of variance explained by each factor. Higher eigenvalues indicate that the factor explains more variance in the original variables.

### Proportion of Variance (Array 2):

- This array shows the proportion of total variance explained by each factor.
- Each value indicates the percentage of total variance in the original variables that is explained by the corresponding factor.

### Cumulative Proportion (Array 3):

- This array provides the cumulative proportion of variance explained as you move through the factors.
- It shows how much of the total variance is explained by the first factor, the first two factors together, the first three factors together, and so on.
- The cumulative proportion helps in deciding how many factors to retain for analysis. Typically, factors with eigenvalues greater than 1 are considered significant, but this criterion can vary depending on the context and specific goals of the analysis.

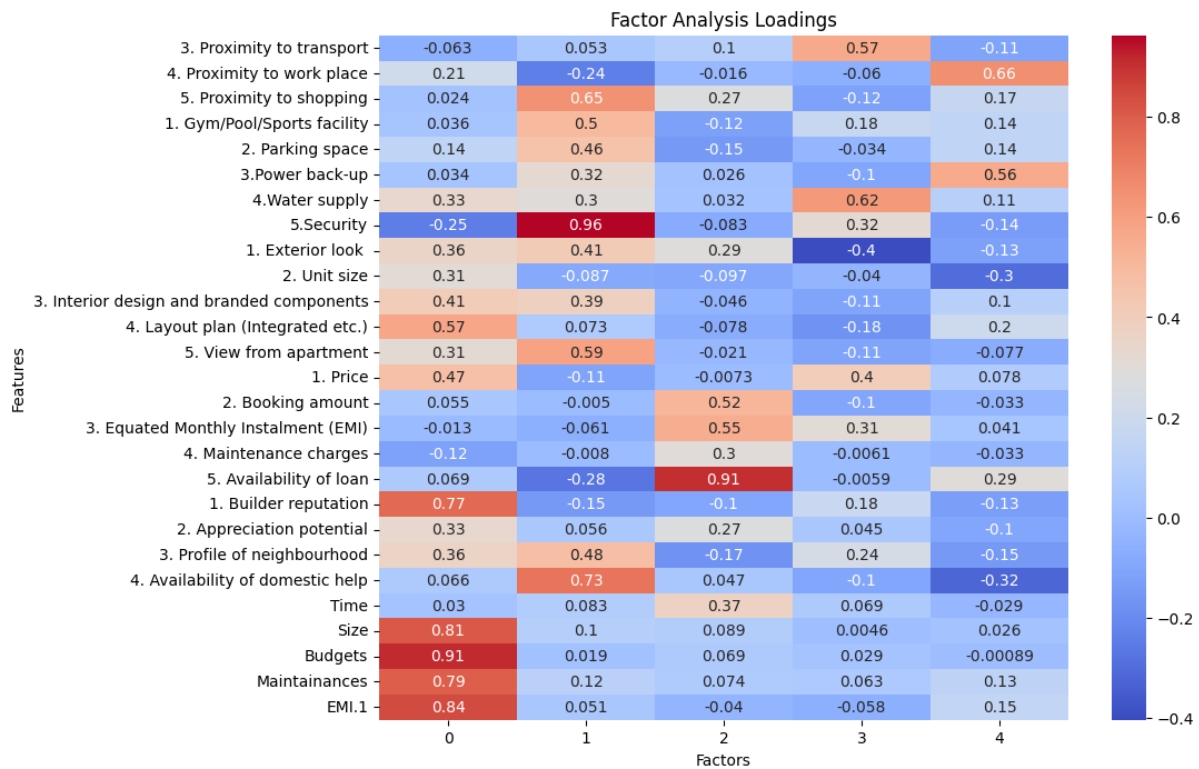
Factor Analysis Loadings:

|                                           | 0         | 1         | 2         | 3 |
|-------------------------------------------|-----------|-----------|-----------|---|
| 3. Proximity to transport                 | -0.062650 | 0.053449  | 0.102691  |   |
| 4. Proximity to work place                | 0.214466  | -0.242324 | -0.015612 |   |
| 5. Proximity to shopping                  | 0.023969  | 0.646406  | 0.273362  |   |
| 1. Gym/Pool/Sports facility               | 0.035508  | 0.497716  | -0.123070 |   |
| 2. Parking space                          | 0.141555  | 0.458574  | -0.152466 |   |
| 3. Power back-up                          | 0.033851  | 0.324156  | 0.025881  |   |
| 4. Water supply                           | 0.334492  | 0.295378  | 0.032303  |   |
| 5. Security                               | -0.247295 | 0.963297  | -0.083131 |   |
| 1. Exterior look                          | 0.357562  | 0.413706  | 0.286870  |   |
| 2. Unit size                              | 0.307274  | -0.086644 | -0.096613 |   |
| 3. Interior design and branded components | 0.408981  | 0.385738  | -0.046064 |   |
| 4. Layout plan (Integrated etc.)          | 0.574897  | 0.072914  | -0.077908 |   |
| 5. View from apartment                    | 0.312396  | 0.592523  | -0.021110 |   |
| 1. Price                                  | 0.468356  | -0.108705 | -0.007293 |   |
| 2. Booking amount                         | 0.055089  | -0.005007 | 0.517348  |   |
| 3. Equated Monthly Instalment (EMI)       | -0.013245 | -0.060950 | 0.550510  |   |
| 4. Maintenance charges                    | -0.118321 | -0.007978 | 0.300754  |   |
| 5. Availability of loan                   | 0.069242  | -0.275819 | 0.909565  |   |
| 1. Builder reputation                     | 0.770507  | -0.151793 | -0.101498 |   |
| 2. Appreciation potential                 | 0.333179  | 0.056012  | 0.270933  |   |
| 3. Profile of neighbourhood               | 0.363382  | 0.482280  | -0.173209 |   |

|                                  |          |          |           |
|----------------------------------|----------|----------|-----------|
| 4. Availability of domestic help | 0.066108 | 0.733663 | 0.046809  |
| Time                             | 0.029564 | 0.083021 | 0.374954  |
| Size                             | 0.809970 | 0.100932 | 0.088567  |
| Budgets                          | 0.914751 | 0.019184 | 0.068697  |
| Maintainances                    | 0.792661 | 0.116358 | 0.073753  |
| EMI.1                            | 0.843329 | 0.050710 | -0.039784 |

|                                           |           |           |
|-------------------------------------------|-----------|-----------|
|                                           | 3         | 4         |
| 3. Proximity to transport                 | 0.566766  | -0.108953 |
| 4. Proximity to work place                | -0.059643 | 0.657572  |
| 5. Proximity to shopping                  | -0.118425 | 0.170156  |
| 1. Gym/Pool/Sports facility               | 0.175301  | 0.143681  |
| 2. Parking space                          | -0.034074 | 0.144277  |
| 3. Power back-up                          | -0.100512 | 0.558963  |
| 4. Water supply                           | 0.617495  | 0.106283  |
| 5. Security                               | 0.319567  | -0.138278 |
| 1. Exterior look                          | -0.404235 | -0.126286 |
| 2. Unit size                              | -0.039999 | -0.298122 |
| 3. Interior design and branded components | -0.109582 | 0.103988  |
| 4. Layout plan (Integrated etc.)          | -0.175612 | 0.204097  |
| 5. View from apartment                    | -0.110427 | -0.077293 |
| 1. Price                                  | 0.403942  | 0.077952  |
| 2. Booking amount                         | -0.102740 | -0.032806 |
| 3. Equated Monthly Instalment (EMI)       | 0.308988  | 0.040785  |
| 4. Maintenance charges                    | -0.006142 | -0.032559 |
| 5. Availability of loan                   | -0.005900 | 0.293333  |
| 1. Builder reputation                     | 0.180912  | -0.127029 |
| 2. Appreciation potential                 | 0.044642  | -0.101233 |
| 3. Profile of neighbourhood               | 0.240515  | -0.148348 |
| 4. Availability of domestic help          | -0.101157 | -0.323017 |
| Time                                      | 0.069180  | -0.028749 |
| Size                                      | 0.004628  | 0.025844  |
| Budgets                                   | 0.028929  | -0.000888 |
| Maintainances                             | 0.063025  | 0.133921  |
| EMI.1                                     | -0.058241 | 0.154306  |

- ☐ The loading value indicates the strength and direction of the relationship between the variable and the factor.
- ☐ Higher absolute values (closer to 1) indicate a stronger relationship between the variable and the factor.
- ☐ Positive values indicate a positive correlation, while negative values indicate a negative correlation.
- ☐ Variables with higher loading values on a particular factor are more strongly influenced by that factor.



This heatmap visualization helps in visually assessing the strength and direction of relationships between variables and factors extracted from the factor analysis.

## PART 2: Cluster Analysis

### INTRODUCTION

In this analysis, we conduct K-means clustering on a dataset related to housing preferences. The dataset includes various factors such as proximity to amenities, property features, financial considerations, and demographic information of potential homebuyers. The aim is to group respondents into distinct clusters based on similarities in their housing preferences and related factors.

### OBJECTIVE

**Identify Consumer Segments:** Utilize K-means clustering to identify distinct segments or clusters of potential homebuyers based on their housing preferences and related factors.

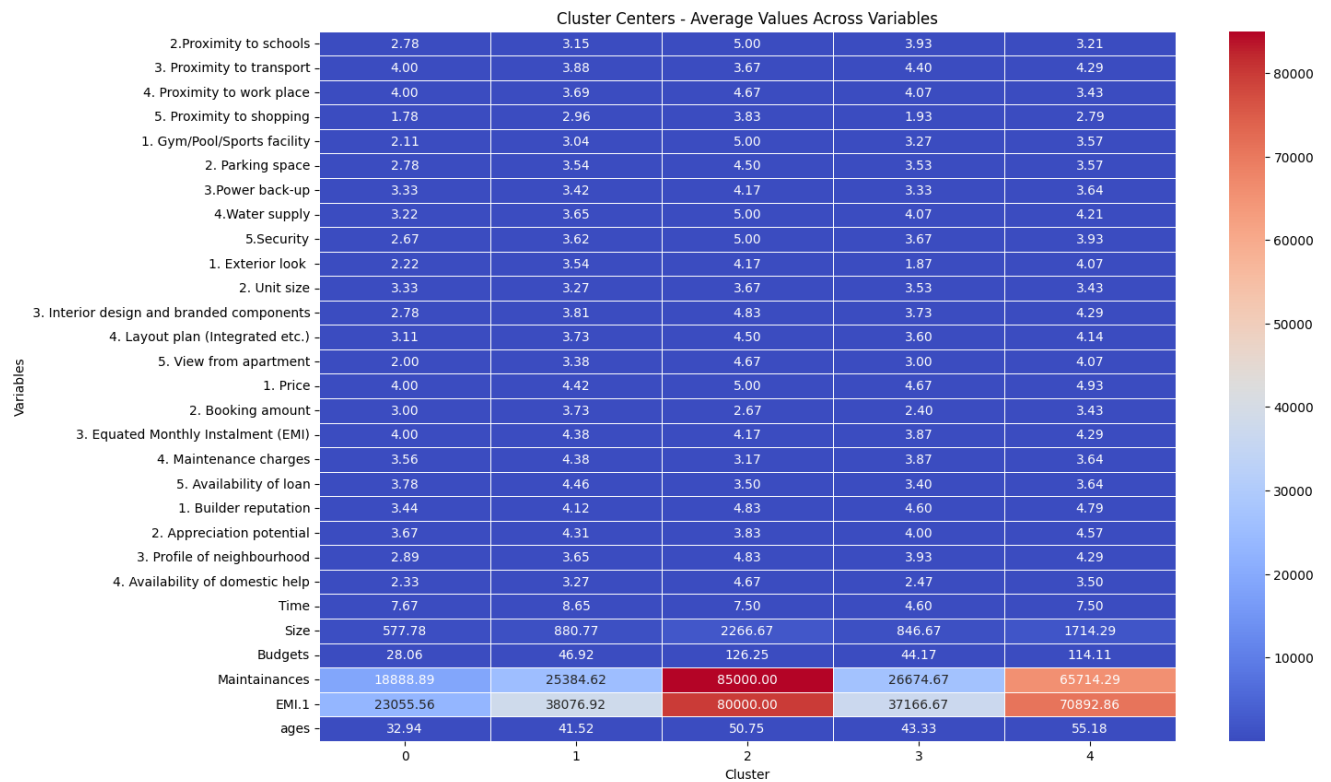
### RESULTS & INTERPRETATIONS

#### Python

Here, we extract the cluster labels, compute the average values for each cluster across the specified columns (from 18 to 46), and then format the data as a matrix suitable for interpretation

|         | 2.Proximity<br>to schools | 3. Proximity<br>to<br>transport | 4. Proximity<br>to work<br>place | 5. Proximity<br>to<br>shopping | 1. Gym/Pool/Sports<br>facility | 2. Parking<br>space | 3.Power<br>back-up | 4.Water<br>supply | 5.Security | 1. Exterior<br>look | ... | 1. Builder<br>reputation | Appreciation<br>potential | 3. Profile of<br>neighbourhood | 4. Availability<br>of domestic<br>help |     |
|---------|---------------------------|---------------------------------|----------------------------------|--------------------------------|--------------------------------|---------------------|--------------------|-------------------|------------|---------------------|-----|--------------------------|---------------------------|--------------------------------|----------------------------------------|-----|
| Cluster |                           |                                 |                                  |                                |                                |                     |                    |                   |            |                     |     |                          |                           |                                |                                        |     |
| 0       | 2.777778                  | 4.000000                        | 4.000000                         | 1.777778                       | 2.111111                       | 2.777778            | 3.333333           | 3.222222          | 2.666667   | 2.222222            | ... | 3.444444                 | 3.666667                  | 2.888889                       | 2.333333                               | 7.6 |
| 1       | 3.153846                  | 3.884615                        | 3.692308                         | 2.961538                       | 3.038462                       | 3.538462            | 3.423077           | 3.653846          | 3.615385   | 3.538462            | ... | 4.115385                 | 4.307692                  | 3.653846                       | 3.269231                               | 8.6 |
| 2       | 5.000000                  | 3.666667                        | 4.666667                         | 3.833333                       | 5.000000                       | 4.500000            | 4.166667           | 5.000000          | 5.000000   | 4.166667            | ... | 4.833333                 | 3.833333                  | 4.833333                       | 4.666667                               | 7.5 |
| 3       | 3.933333                  | 4.400000                        | 4.066667                         | 1.933333                       | 3.266667                       | 3.533333            | 3.333333           | 4.066667          | 3.666667   | 1.866667            | ... | 4.600000                 | 4.000000                  | 3.933333                       | 2.466667                               | 4.6 |
| 4       | 3.214286                  | 4.285714                        | 3.428571                         | 2.785714                       | 3.571429                       | 3.571429            | 3.642857           | 4.214286          | 3.928571   | 4.071429            | ... | 4.785714                 | 4.571429                  | 4.285714                       | 3.500000                               | 7.5 |

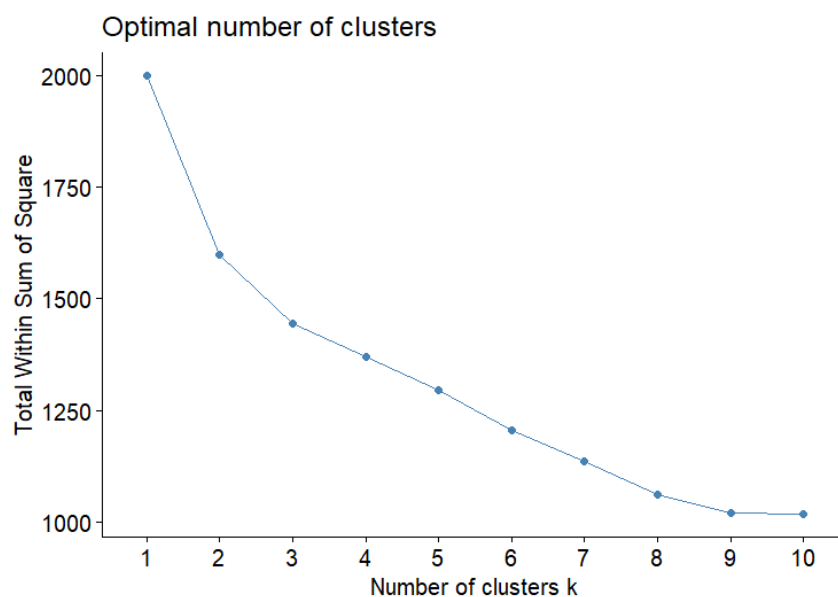
5 rows x 29 columns



The five clusters represent distinct groups of respondents with similar characteristics and preferences.

This clustering analysis helps to segment the respondents based on their preferences and characteristics. By examining the average values of these clusters, we can identify distinct groups with similar attributes and preferences, which can be used for targeted marketing, policy making, or further analysis.

## R



K-means clustering with 5 clusters of sizes 20, 12, 26, 6, 6

Cluster means:

X1.Proximity.to.city X2.Proximity.to.schools

|   |             |             |
|---|-------------|-------------|
| 1 | 0.42645297  | -0.23895587 |
| 2 | 0.04373877  | 0.30220889  |
| 3 | -0.05929967 | -0.09514985 |
| 4 | -1.86983224 | -0.92771101 |
| 5 | 0.61781007  | 1.53212878  |

X3..Proximity.to.transport

|   |             |
|---|-------------|
| 1 | 0.03920684  |
| 2 | 0.35939603  |
| 3 | -0.09801710 |
| 4 | 0.13068947  |
| 5 | -0.55543023 |

X4..Proximity.to.work.place

|   |             |
|---|-------------|
| 1 | -0.41689837 |
| 2 | 0.60892428  |
| 3 | 0.04431367  |
| 4 | -0.89443650 |
| 5 | 0.87422325  |

X5..Proximity.to.shopping

|   |            |
|---|------------|
| 1 | 0.1550084  |
| 2 | -1.0151533 |
| 3 | 0.2777527  |
| 4 | -1.2279100 |
| 5 | 1.5379269  |

X1..Gym.Pool.Sports.facility X2..Parking.space

|   |            |             |
|---|------------|-------------|
| 1 | 0.4028055  | 0.24623856  |
| 2 | -0.3608466 | -0.28043836 |
| 3 | -0.4173297 | -0.04103976 |
| 4 | -0.3608466 | -1.47743137 |
| 5 | 1.5482837  | 1.39535185  |

X3.Power.back.up X4.Water.supply X5.Security

|   |            |             |            |
|---|------------|-------------|------------|
| 1 | 0.1644957  | 0.12688813  | 0.3133435  |
| 2 | -0.6853986 | 0.00352467  | -0.7141973 |
| 3 | 0.1265351  | -0.32860771 | -0.1289855 |
| 4 | -0.8224783 | -0.61329261 | -0.6153953 |
| 5 | 1.0966378  | 1.60724959  | 1.5582487  |

X1..Exterior.look X2..Unit.size

|   |            |             |
|---|------------|-------------|
| 1 | 0.6750830  | 0.07483749  |
| 2 | -1.0223240 | 0.27440412  |
| 3 | 0.1110216  | -0.25329611 |
| 4 | -1.4975979 | -0.09978332 |
| 5 | 0.8108756  | 0.39913326  |

X3..Interior.design.and.branded.components

|   |            |
|---|------------|
| 1 | 0.4509889  |
| 2 | -0.6434109 |
| 3 | 0.1012991  |
| 4 | -1.9061799 |
| 5 | 1.2507427  |

X4..Layout.plan..Integrated.etc..

|   |            |
|---|------------|
| 1 | 0.4194346  |
| 2 | -0.2401111 |
| 3 | -0.1010010 |
| 4 | -1.4102729 |
| 5 | 0.9300507  |

X5..View.from.apartment X1..Price

|   |             |           |
|---|-------------|-----------|
| 1 | 0.72152166  | 0.3225928 |
| 2 | -1.03925597 | 0.3225928 |

```

3      -0.02342272 -0.4069632
4      -1.42539141 -0.7312104
5        1.20032961  0.7742227
X2..Booking.amount
1        0.1941775
2       -0.7605287
3        0.3659500
4       -0.1941775
5       -0.5178068
X3..Equated.Monthly.Instalment..EMI.
1          -0.2623140
2          -0.2623140
3          0.2266145
4          0.4439160
5          -0.0269040
X4..Maintenance.charges
1        0.1794032
2       -0.5476518
3        0.3675223
4       -0.1070124
5       -0.9882911
X5..Availability.of.loan X1..Builder.reputation
1       -0.4883082      0.6235284
2       -0.0813847      0.2267376
3        0.7324623     -0.4854510
4       -0.8952317     -1.0958985
5       -0.4883082      0.6676163
X2..Appreciation.potential
1        0.6989383
2       -0.1436707
3       -0.1541249
4       -0.8231940
5       -0.5513847
X3..Profile.of.neighbourhood
1        0.6393887
2       -0.3629863
3       -0.4257468
4       -0.9457625
5        1.3853422
X4..Availability.of.domestic.help      Time
1        0.41450276 -0.11583378
2       -1.07867677 -0.31604031
3       -0.06712595  0.31153016
4       -0.48479855 -0.36609194
5        1.55135536  0.03432112
      Size  Budgets Maintainances  EMI.1
1  0.6695344  0.8396597  0.6873455  0.8241764
2 -0.4038462 -0.2958008 -0.4643225 -0.2718113
3 -0.4549401 -0.5686006 -0.4818407 -0.4857878
4 -1.2806175 -1.2667166 -1.0693715 -1.6070247
5  1.8279353  1.5233888  1.7948410  1.5084733

```

Clustering vector:

```

[1] 2 2 4 5 3 2 1 4 2 1 3 2 3 3 2 4 2 2 1 1 2 3 2
[24] 3 5 3 3 3 2 1 1 3 3 1 4 1 3 1 3 3 3 1 1 3 3 1
[47] 3 3 5 4 4 5 3 5 1 3 3 5 1 3 1 1 1 1 2 3 1 3 3
[70] 1

```

Within cluster sum of squares by cluster:

```

[1] 368.91294 189.29228 446.08704 171.58257

```



```
[5] 83.50496
(between_SS / total_SS = 37.1 %)

Available components:

[1] "cluster"    "centers"    "totss"
[4] "withinss"   "tot.withinss" "betweenss"
[7] "size"       "iter"       "ifault"
```

**Cluster Sizes:** The sizes of the clusters are given as 6, 20, 18, 20, and 6, indicating the number of data points in each cluster.

**Cluster Means (Centroids):** These are the mean values of each feature (or dimension) within each cluster.

**Clustering Vector:** This vector shows the cluster assignment for each data point in the dataset. For instance:

- Data point 1 belongs to Cluster 2, data point 2 to Cluster 2, and so on.
- It provides a way to map each data point back to its respective cluster after clustering.

**Within Cluster Sum of Squares:** This metric measures the sum of squares of distances between data points and their cluster centroids within each cluster. Lower values indicate tighter clusters where data points are closer to their centroids.

**Percentage of Variance Explained:** The percentage of variance explained by the clustering solution is also provided. In this case, it's 37%, which tells how well the clusters separate the data points based on the features used in clustering.

**Available Components:** These are the components available from the clustering result, including cluster assignments, centroids, total sum of squares (total variance in the data), within cluster sum of squares, between cluster sum of squares, cluster sizes, iteration count, and potential error codes (ifault).

## PART 3: Multidimensional Scaling

### INTRODUCTION

In this analysis, we aim to explore the multidimensional characteristics of various ice cream brands based on their attributes.

### OBJECTIVE

Use clustering techniques to group ice cream brands based on similarities in their attribute profiles, aiming to identify distinct segments of brands with similar characteristics.

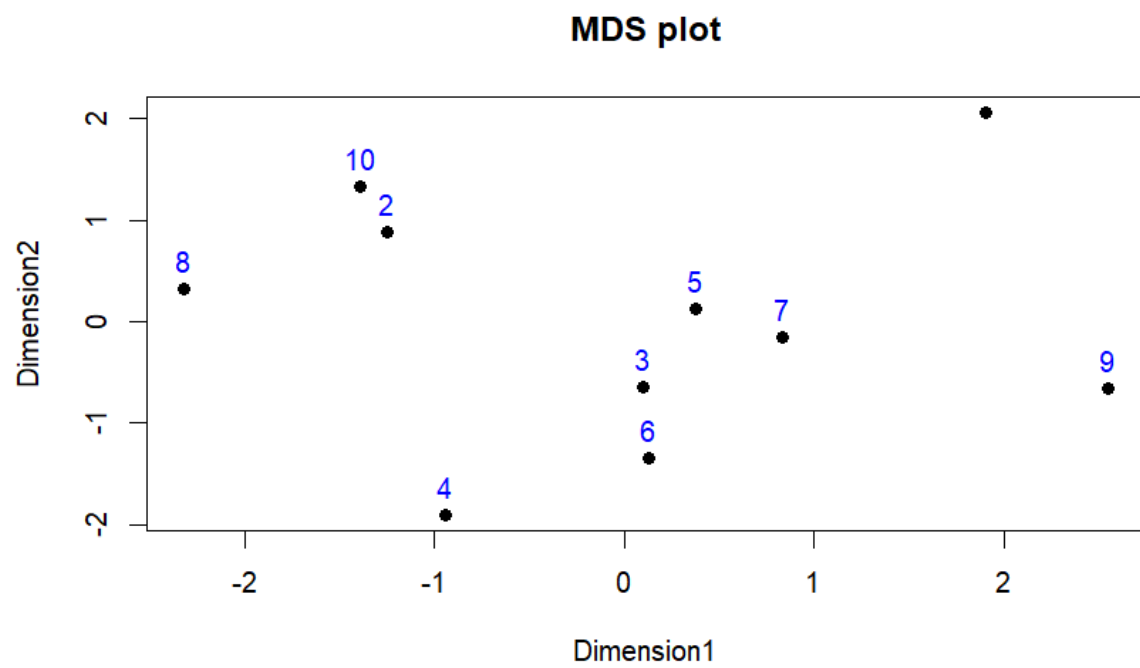
### RESULTS & INTERPRETATIONS

#### R

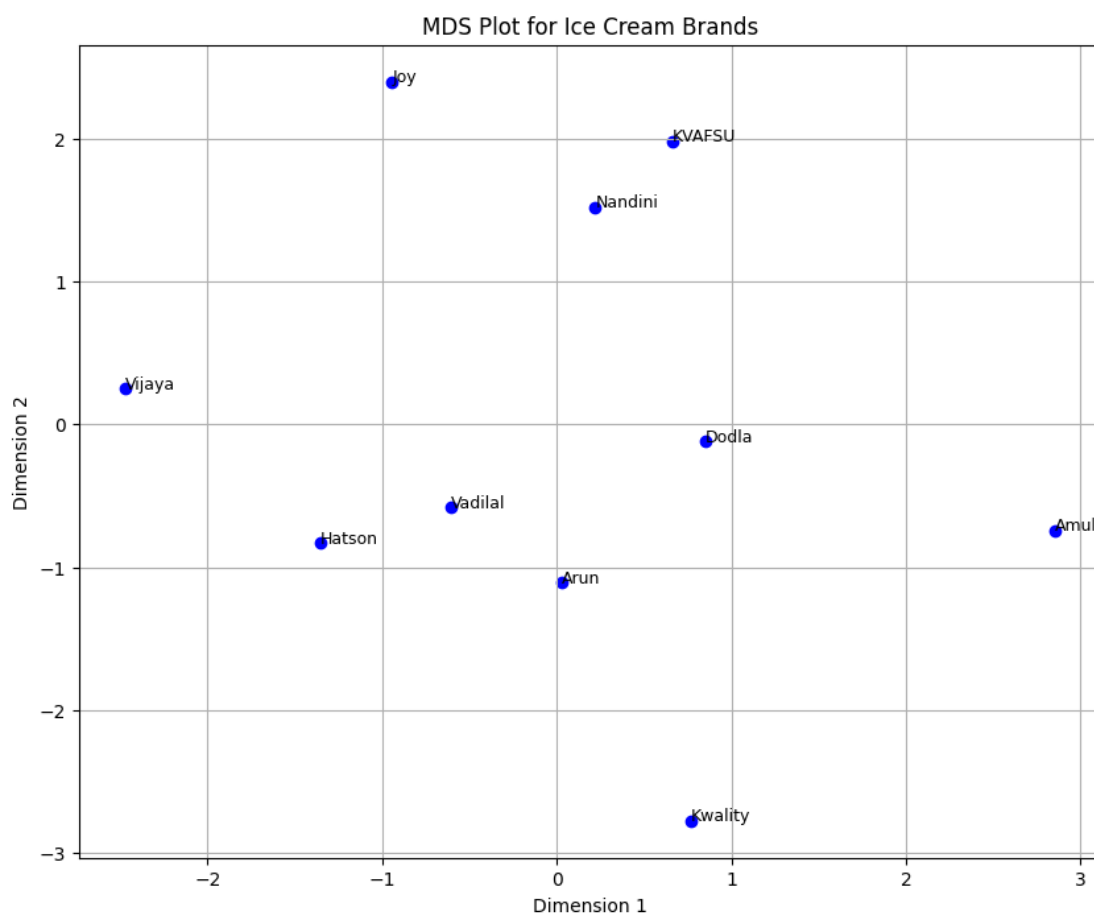
```
> (icecream_df)
```

|    | Brand   | Price | Availability | Taste | Flavour | Consistency | Shelflife |
|----|---------|-------|--------------|-------|---------|-------------|-----------|
| 1  | Amul    | 4     | 5            | 4     | 3       | 4           | 3         |
| 2  | Nandini | 3     | 2            | 3     | 2       | 3           | 3         |
| 3  | Vadilal | 2     | 2            | 4     | 3       | 4           | 4         |
| 4  | Vijaya  | 3     | 1            | 3     | 5       | 3           | 4         |
| 5  | Dodla   | 3     | 3            | 3     | 4       | 4           | 3         |
| 6  | Hatson  | 2     | 2            | 4     | 4       | 3           | 4         |
| 7  | Arun    | 2     | 3            | 4     | 3       | 4           | 4         |
| 8  | Joy     | 4     | 1            | 2     | 3       | 3           | 3         |
| 9  | Kwality | 3     | 4            | 5     | 5       | 4           | 4         |
| 10 | KVAFSU  | 4     | 2            | 3     | 2       | 3           | 3         |

- In the MDS plot, closer points represent ice cream brands that are more similar based on their attributes (Price, Availability, Taste, Flavour, Consistency, Shelflife)
- Brands that are further apart in the plot are more dissimilar in terms of their attribute profiles.
- The plot provides a visual representation of how ice cream brands cluster or spread out in a 2-dimensional space based on their attribute similarities.



## Python



## PART 4: Conjoint Analysis

### INTRODUCTION

In this Conjoint Analysis study using a pizza dataset, we aim to understand consumer preferences and decision-making processes when selecting pizzas based on various attributes such as brand, price, weight, crust type, cheese, size, toppings, and spiciness. Conjoint Analysis allows us to determine the relative importance of each attribute and how different levels of these attributes influence consumer choices.

### OBJECTIVE

**Attribute Importance:** Assess the relative importance of each pizza attribute (brand, price, weight, crust, cheese, size, toppings, spiciness) in influencing consumer preferences.

### RESULTS & INTERPRETATIONS

#### Python

| OLS Regression Results |                  |                     |          |       |        |        |
|------------------------|------------------|---------------------|----------|-------|--------|--------|
| =====                  |                  |                     |          |       |        |        |
| Dep. Variable:         | y                | R-squared:          | 0.989    |       |        |        |
| Model:                 | OLS              | Adj. R-squared:     | 0.977    |       |        |        |
| Method:                | Least Squares    | F-statistic:        | 81.76    |       |        |        |
| Date:                  | Mon, 08 Jul 2024 | Prob (F-statistic): | 3.19e-06 |       |        |        |
| Time:                  | 17:13:32         | Log-Likelihood:     | -10.770  |       |        |        |
| No. Observations:      | 16               | AIC:                | 39.54    |       |        |        |
| Df Residuals:          | 7                | BIC:                | 46.49    |       |        |        |
| Df Model:              | 8                |                     |          |       |        |        |
| Covariance Type:       | nonrobust        |                     |          |       |        |        |
| =====                  |                  |                     |          |       |        |        |
|                        | coef             | std err             | t        | P> t  | [0.025 | 0.975] |
| -----                  |                  |                     |          |       |        |        |
| const                  | 17.5500          | 0.605               | 28.992   | 0.000 | 16.119 | 18.981 |
| x1                     | 0.0500           | 0.160               | 0.312    | 0.764 | -0.329 | 0.429  |
| x2                     | -0.4500          | 0.160               | -2.806   | 0.026 | -0.829 | -0.071 |
| x3                     | -3.5500          | 0.160               | -22.138  | 0.000 | -3.929 | -3.171 |
| x4                     | -3.5000          | 0.359               | -9.761   | 0.000 | -4.348 | -2.652 |
| x5                     | 0.5000           | 0.359               | 1.394    | 0.206 | -0.348 | 1.348  |
| x6                     | 0.5000           | 0.359               | 1.394    | 0.206 | -0.348 | 1.348  |
| x7                     | -2.2500          | 0.359               | -6.275   | 0.000 | -3.098 | -1.402 |
| x8                     | -1.5000          | 0.359               | -4.183   | 0.004 | -2.348 | -0.652 |
| =====                  |                  |                     |          |       |        |        |
| Omnibus:               | 6.427            | Durbin-Watson:      | 1.387    |       |        |        |
| Prob(Omnibus):         | 0.040            | Jarque-Bera (JB):   | 1.576    |       |        |        |
| Skew:                  | 0.009            | Prob(JB):           | 0.455    |       |        |        |
| Kurtosis:              | 1.463            | Cond. No.           | 11.7     |       |        |        |
| =====                  |                  |                     |          |       |        |        |

Based on the coefficients:

- **Negative Coefficients** :These indicate that as the level of the attribute increases from the baseline, the preference or ranking decreases. For example, if x2 represents a higher price level, a negative coefficient suggests that higher prices are less preferred.
- **Positive Coefficients**: These indicate that as the level of the attribute increases from the baseline, the preference or ranking increases. For example, if x1 represents a specific feature that consumers prefer, a positive coefficient suggests higher preference for that feature.

## R

```
Call:
lm(formula = ranking ~ price + size + crust + cheese + spicy,
    data = pizza)

Residuals:
    Min       1Q   Median       3Q      Max
-7.000 -3.000  0.250  3.375  6.500

Coefficients:
              Estimate Std. Error t value
(Intercept)      11.250      4.149   2.711
price$2.00        -0.750      4.149  -0.181
price$3.00        -0.750      4.149  -0.181
price$4.00       -1.500      4.149  -0.361
sizeregular        0.500      2.934   0.170
crustthin        -3.500      2.934  -1.193
cheeseMozzarella   0.500      2.934   0.170
spicynormal      -1.500      2.934  -0.511

              Pr(>|t|)
(Intercept)    0.0266 *
price$2.00     0.8611
price$3.00     0.8611
price$4.00     0.7271
sizeregular    0.8689
crustthin      0.2671
cheeseMozzarella 0.8689
spicynormal    0.6230
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
  ' ' 1

Residual standard error: 5.868 on 8 degrees of freedom
Multiple R-squared:  0.1897,    Adjusted R-squared:  -0.5193
F-statistic: 0.2676 on 7 and 8 DF,  p-value: 0.9504
```

## **Business Significance**

FA helps businesses understand underlying factors influencing customer decisions, such as housing preferences. By identifying these factors, businesses can tailor marketing strategies, product offerings, and customer experiences to better meet customer needs and preferences.

PCA is crucial for businesses to simplify complex datasets and understand the most significant attributes driving customer preferences. This insight helps in product development, marketing strategies, and customer segmentation.

MDS helps businesses understand how products or brands are perceived relative to each other by consumers. This insight is valuable for competitive analysis, brand positioning, and identifying market segments based on consumer preferences.

Conjoint Analysis provides insights into product features that drive customer choices and willingness to pay. Businesses can use these insights for pricing strategies, product development, and market positioning to meet customer expectations effectively.

These analyses collectively help businesses gain deep insights into consumer behavior, preferences, and decision-making processes, enabling them to make data-driven decisions that align with customer needs and enhance business performance.